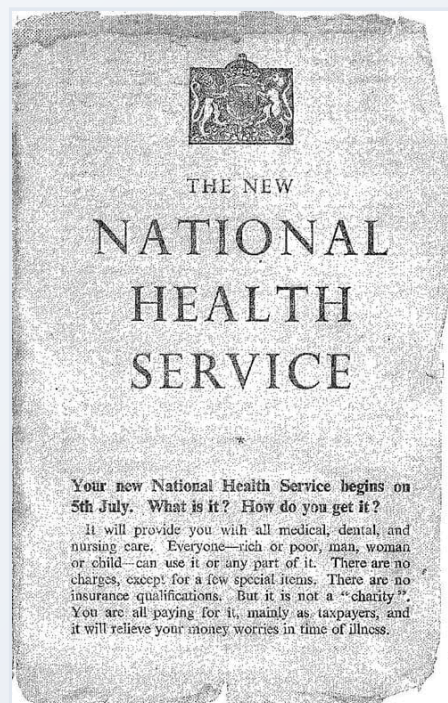


MODERN

Paper 1: Medicine Through Time with the Western Front



COURSE TEXTBOOK

Enquiry: How has medicine changed from c1250 to the modern day?

Contents

Lesson 1: KT4.1: How have modern discoveries changed our understanding of illness?

Lesson 2: KT4.2: How have prevention and treatment advanced in the modern era?

Lesson 3: KT4.3: How was Penicillin discovered and mass-produced?

Lesson 4: KT4.4: How has the government tackled the modern epidemic of Lung Cancer?

L1: KT4.1: How have modern discoveries changed our understanding of illness?

LEARNING OBJECTIVES

- ☐ Explain how the discovery of DNA and the Human Genome Project transformed the understanding of genetic and hereditary illnesses.
- ☐ Analyze the influence of lifestyle choices on modern health and the impact of technology on medical diagnosis.

The 20th century witnessed a monumental shift in medicine. With Germ Theory firmly established, scientists soon realised that microbes did not cause every single illness. The groundbreaking discovery of DNA finally unlocked the mystery of hereditary diseases, whilst researchers proved that our own modern lifestyle choices were causing new killer illnesses like cancer. Simultaneously, a technological revolution in diagnosis completely transformed how doctors accurately identified what was wrong with their patients.

Did you know?

- 1953: The exact year James Watson and Francis Crick successfully discovered the double-helix structure of DNA.
- 1990: The launch year of the Human Genome Project, a massive global scientific effort to decode and map all human DNA.
- 1950: The year the British Medical Research Council published a conclusive study firmly linking smoking cigarettes to lung cancer.

Recall & Retrieval

- | | |
|---|---|
| 1. What disease did Edward Jenner vaccinate against in 1796? | 2. What did Edward Jenner observe that led to his vaccine? |
| 3. Who discovered the cause of cholera in 1854? | 4. Which Roman physician developed the Theory of Opposites? |
| 5. What was the Royal Society? | 6. Who introduced antiseptic surgery using carbolic acid in 1865? |
| 7. What did Harvey prove wrong about Galen's theories of blood? | 8. Who published the Germ Theory in 1861? |
| 9. What invention helped spread new medical ideas during the Renaissance? | 10. What was 'transference'? |

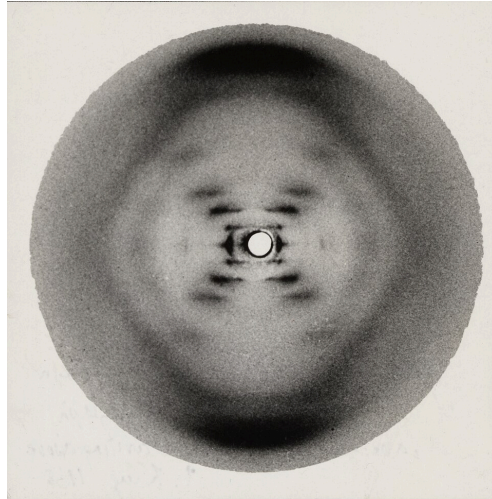
Vocabulary Check

The Odd One Out: Circle *ONE* term that does not belong with the others. Explain your reasoning: What historical connection links the other terms, and why is your chosen term different?

DNA | Rosalind Franklin | Human Genome Project | Biopsy | Electron Microscope | Inherited Diseases

Teacher Model / Pupil Reasoning:

Source S1 : Rosalind Franklin's Photograph 51 (1952)



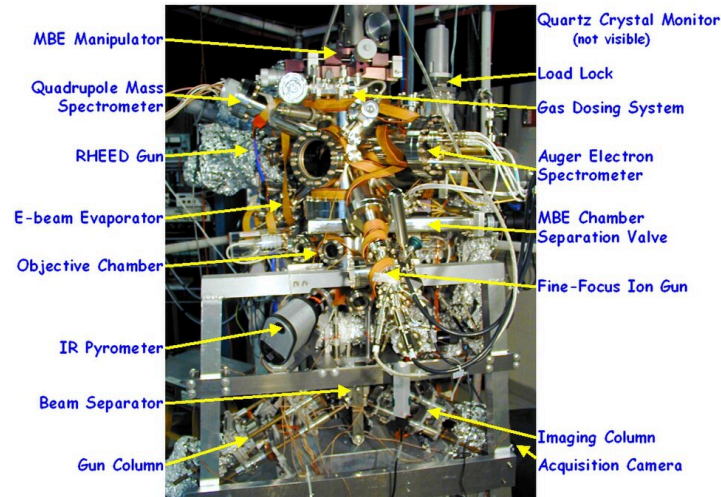
Photograph 51: the historic X-ray crystallography image that provided the critical mathematical proof of the double helix structure of DNA.

Source S2 : Watson and Crick with the Wire DNA Model (1953)



James Watson and Francis Crick admiring their full-scale molecular wire model of the DNA double helix at the Cavendish Laboratory.

Source S3 : Transmission Electron Microscope (TEM) Imagery of Virus



High-resolution transmission electron micrograph showing individual virus particles magnified over 100,000 times.

DNA and Modern Genetics By 1900, doctors knew that microbes caused infectious diseases, but they still couldn't explain hereditary diseases like Haemophilia. In 1953, James Watson and Francis Crick, using X-ray photographs taken by Rosalind Franklin, discovered the double-helix structure of DNA. This was a monumental breakthrough because it proved how genetic information was passed from parents to children. In 1990, scientists launched the Human Genome Project, an international effort to map every single gene in the human body. They published a working draft of the genome in 2000, and the project was essentially complete by 2003. This incredible achievement allowed scientists to identify the specific genes responsible for diseases like breast cancer and cystic fibrosis, opening the door to genetic screening and personalized medicine.

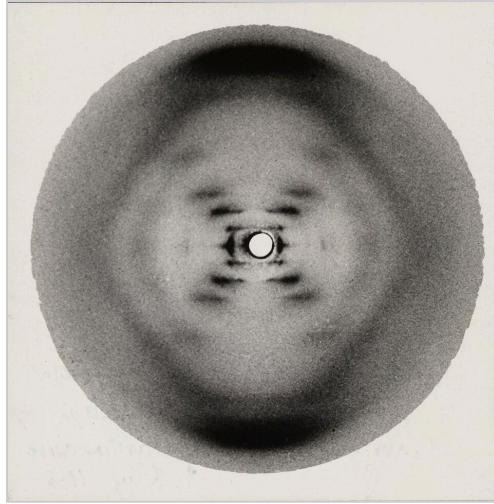
James Watson & Francis Crick — Molecular Biologists

Working together at Cambridge University, they successfully published the double-helix structure of DNA in 1953. Their monumental breakthrough finally allowed scientists to understand how genetic traits and hereditary diseases are passed down from parents to children.

Key actions: Published the first accurate molecular model of the double-helix structure of DNA in 1953, utilizing X-ray diffraction data and physical cardboard models to solve the genetic puzzle.

Significance: Unlocked the genetic paradigm of modern medicine, explaining how hereditary characteristics are passed down and directly enabling the launch of the Human Genome Project (1990-2003).

Limitations: Their discovery was purely theoretical. Understanding the physical structure of DNA did not translate into immediate cures; it took decades of research to link genes to specific diseases like lung cancer, and medical science still lacks the technology to safely and universally edit genetic mutations today.



Source A: Source A: Rosalind Franklin's Photograph 51 (1952)

Rosalind Franklin — X-ray Crystallographer

A brilliant scientist whose exceptionally clear X-ray photographs of DNA (specifically "Photograph 51") provided the crucial, undeniable evidence of the double-helix shape. Her work was shared with Watson and Crick without her explicit permission.

Key actions: Captured Photo 51 (1952), a highly precise X-ray crystallography photograph of crystallized DNA fibers, and calculated the physical dimensions of the helix.

Significance: Provided the essential, mathematical proof of DNA's double-helix structure, which Watson and Crick relied upon to build their successful molecular model.

Limitations: Her impact was severely limited by systemic sexism in the 1950s academic community; her data was shared without her permission, and she was denied co-authorship on the initial discovery paper. Because her expertise was strictly in chemical physics, she played no role in translating the discovery of DNA into clinical medical therapies.



Source B: Source B: Watson and Crick with the Wire DNA Model (1953)

The Discovery of DNA and Genetics In 1953, scientists James Watson and Francis Crick famously discovered the double-helix structure of DNA at Cambridge University. Their discovery built heavily upon the vital X-ray crystallography image (Photo 51) taken by Rosalind Franklin and Maurice Wilkins in 1951, which provided the exact physical dimensions they needed. This breakthrough finally proved how characteristics and hereditary diseases (illnesses caused by genetic factors) are passed from parents to children. In 1990, the Human Genome Project was launched to decode and map the exact purpose of

every single gene in the human body, a massive global effort that produced a working draft in 2000 and was essentially complete by 2003. Discovering DNA was a monumental breakthrough because it finally allowed scientists to identify the specific genes that cause hereditary conditions (like cystic fibrosis and Down's syndrome), leading to modern genetic testing and preventative treatments.

KNOWLEDGE CHECK (Q1)

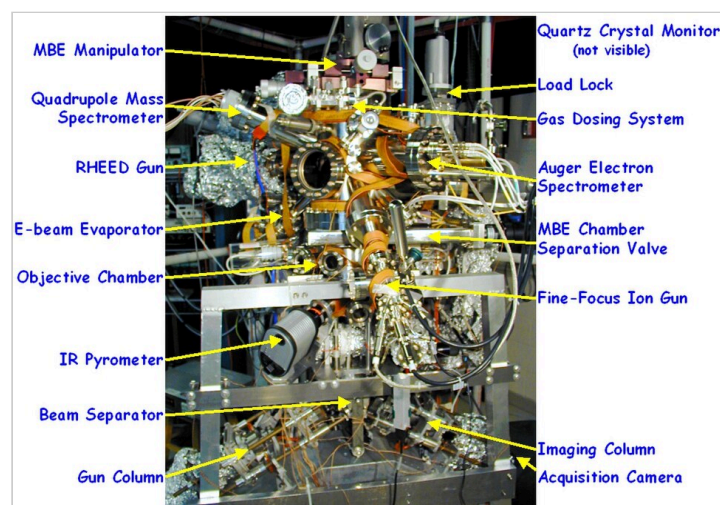
How did Watson and Crick's discovery of the double-helix structure of DNA help scientists search for the causes of hereditary diseases?

- ☐ A. It provided a complete blueprint of human biology, allowing scientists to identify specific faulty genes responsible for conditions like cystic fibrosis.
- ☐ B. It proved that hereditary diseases were caused by viruses.
- ☐ C. It allowed scientists to create antibiotics to kill hereditary diseases.
- ☐ D. It proved that hereditary diseases were caused by miasma.

Q2. Identify who discovered the double-helix structure of DNA in 1953.

The Influence of Lifestyle Factors As infectious diseases declined during the 20th century, doctors realised that modern lifestyle choices were causing a massive rise in new, preventable illnesses. In 1950, the British Medical Research Council published a study conclusively linking smoking to the huge rise in lung cancer. It is also directly linked to high blood pressure, throat cancer, and heart disease. Scientists also proved that a poor diet heavily impacts health: eating too much sugar causes incurable type 2 diabetes, while eating too much fat can lead to heart disease. Other dangerous lifestyle factors identified include binge drinking alcohol (causing liver and kidney diseases) and unprotected sex or sharing needles (spreading diseases like HIV). Understanding lifestyle factors fundamentally changed medicine because it proved that many modern killer diseases are entirely preventable, shifting the government's focus towards public health campaigns and personal responsibility.

Q3. Describe the goal of the Human Genome Project.



Source C: Source C: Transmission Electron Microscope (TEM) Imagery of Virus

Improvements in Diagnosis and Technology At the start of the 20th century, doctors diagnosed patients largely by observing their external symptoms and making a knowledgeable guess. However, modern medicine shifted towards laboratory testing (such as examining skin or tissue samples in a biopsy) and high-tech diagnostic equipment. Blood tests (from the 1930s) now allow doctors to test for an enormous number of conditions using the patient's blood without needing invasive surgery. Advances in imaging include traditional x-rays (1890s) for broken bones, ultrasound scans (1940s) for internal organs, and CT and MRI scans (from the 1970s) which create detailed 3D images to diagnose tumours and soft tissue injuries. Other tools include ECGs (electro cardiograms) to track heart activity and endoscopes (flexible cameras) to see inside the digestive system. This technological revolution meant that diagnosis was no longer based on a doctor's guesswork; instead, doctors could identify the exact microbe, tumour, or internal issue causing the illness, allowing for highly accurate, targeted treatments.

Q4. Explain why Germ Theory could not explain all illnesses.

Q5. Evaluate the impact of the discovery of DNA on modern medicine.

Pair & Share Activity

Q6. Which modern development has had a more significant impact on our understanding of the causes of illness: the discovery of genetic factors (DNA and hereditary disease) or the scientific identification of lifestyle factors (smoking, diet, and alcohol)?

Q7. 'The discovery of DNA was the most significant breakthrough in understanding the causes of illness in the period c1900-present.' How far do you agree? Explain your answer. (16 marks) You may use the following in your answer:

- Watson and Crick
- Lifestyle factors

You must also use information of your own.

GCSE Exam Practice

Q8. Explain why the discovery of the structure of DNA was a significant breakthrough in understanding the causes of illness. (12 marks)

Q9. Explain one way in which ideas about the cause of illness in the modern period (c1900-present) were different from ideas in the period c1700-c1900. (4 marks)

L2: KT4.2: How have prevention and treatment advanced in the modern era?

LEARNING OBJECTIVES

- ☐ Explain how scientific and technological developments led to the creation of magic bullets and antibiotics.
- ☐ Analyze the impact of the National Health Service (NHS) on the provision of and access to medical care.

The 20th century transformed medicine from a system that merely managed symptoms into one that actively destroyed diseases. The discovery of targeted chemical cures and powerful antibiotics saved millions of lives from previously fatal infections. Meanwhile, the creation of the NHS revolutionised access to these high-tech treatments, ensuring that for the first time in history, life-saving healthcare and active prevention campaigns were available to everyone, regardless of their wealth.

Did you know?

- 1909: The year Paul Ehrlich's team discovered Salvarsan 606, the world's very first 'magic bullet' to cure syphilis.
- 1948: The year the National Health Service (NHS) was officially launched, providing free healthcare to all.
- 1956: The year the polio vaccine was introduced in the UK, saving thousands of children from death and paralysis.

Recall & Retrieval

- | | |
|---|--|
| 1. What structure did Watson and Crick discover in 1953? | 2. What is the function of DNA in relation to disease? |
| 3. Name one modern lifestyle factor that causes disease. | 4. Which female scientist's X-ray crystallography photo was vital to Watson and Crick's DNA discovery? |
| 5. Why did the mapping of the human genome not lead to immediate cures? | 6. What did Harvey prove wrong about Galen's theories of blood? |
| 7. Who published the Germ Theory in 1861? | 8. Who introduced antiseptic surgery using carbolic acid in 1865? |
| 9. Who wrote 'On the Fabric of the Human Body' (1543)? | 10. What was the significance of the Public Health Act of 1875? |

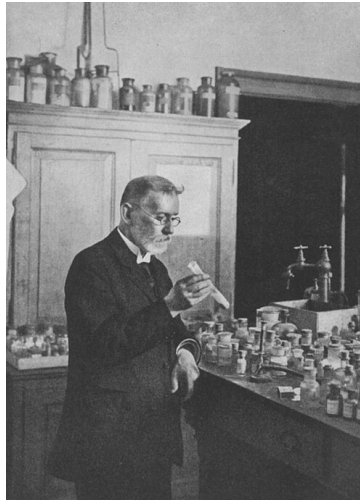
Vocabulary Check

The Golden Sentence: Choose *TWO* terms from the word bank. Write *ONE* grammatically sophisticated, historically accurate sentence connecting them using a causal conjunction (*because, although, or consequently*):

Magic Bullet | Salvarsan 606 | Prontosil | Antibiotics | National Health Service (NHS) | Beveridge Report

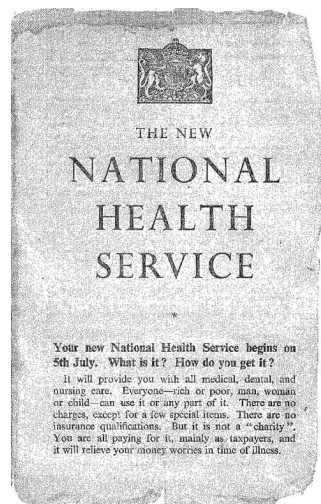
Teacher Model / Pupil Sentence:

Source S1 : Paul Ehrlich in his Frankfurt Laboratory (c.1910)



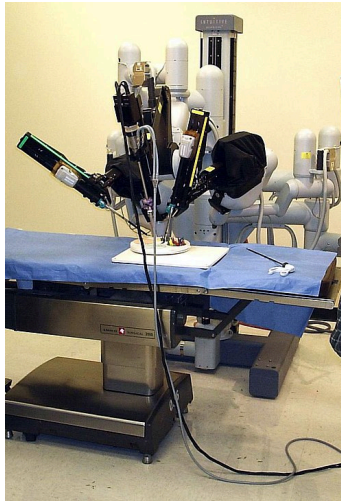
German scientist Paul Ehrlich in his laboratory surrounded by hundreds of chemical compounds, seeking 'magic bullets' to destroy targeted bacteria.

Source S2 : 1948 Ministry of Health NHS Information Leaflet



'The new National Health Service begins on 5th July': the official government pamphlet explaining free medical, dental, and hospital care.

Source S3 : Modern Da Vinci Robotic Keyhole Surgery Unit



A state-of-the-art robotic surgical console allowing surgeons to perform minimally invasive laparoscopic keyhole operations with microscopic precision.

High-Tech Diagnostics and Magic Bullets In the 20th century, diagnosing illness shifted from observing external symptoms to looking inside the body. X-rays, discovered by Wilhelm Roentgen in 1895, were joined by Marie Curie's pioneering research into radioactivity, which she used to build mobile X-ray units ('petites Curies') that treated wounded soldiers in the First World War. Modern technology later introduced Ultrasound, MRI, and CT scans. Doctors no longer had to cut a patient open to see what was wrong. Treatment also evolved rapidly. In 1909, Paul Ehrlich discovered Salvarsan 606, the first 'Magic Bullet'-a chemical compound that targeted and killed the specific bacteria causing syphilis without harming the rest of the body. Later, in 1932, Gerhard Domagk discovered the second magic bullet, Prontosil, which cured blood poisoning. These synthetic chemicals marked the beginning of modern pharmaceutical treatments.

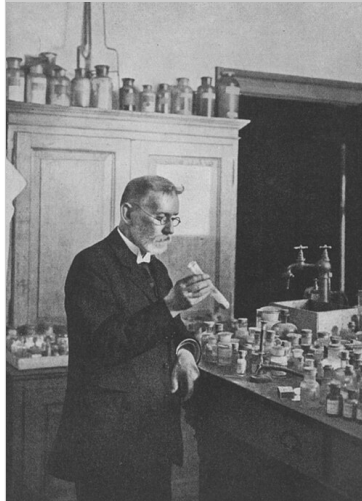
Marie Curie — Physicist and Chemist

A pioneering scientist who discovered radium and polonium. During WWI, she equipped the French army with 20 mobile X-ray units (known as "Petites Curies") so that wounded soldiers could be diagnosed directly on the frontline.

Key actions: Polish-French physicist who realized the urgent military need for radiology on the Western Front; equipped 20 mobile X-ray vans ("petites Curies") and trained 150 women to operate them.

Significance: Revolutionized battlefield triage by bringing diagnostic X-ray technology directly to forward Casualty Clearing Stations (CCS), allowing frontline surgeons to rapidly locate shrapnel and bullets before gas gangrene set in.

Limitations: Her mobile units were exceptionally fragile, required constant engine power from the vans to run, and struggled to navigate the destroyed, mud-choked terrain of the Ypres Salient or the Somme. Furthermore, a lack of protective shielding meant she and her operators suffered chronic, lethal exposure to radiation.



Source A: Source A: Paul Ehrlich in his Frankfurt Laboratory (c.1910)

Paul Ehrlich — German Physician

A former student of Robert Koch who theorized the existence of "magic bullets" - chemical compounds that could target and destroy specific disease microbes inside the body without harming the rest of the patient. He co-discovered Salvarsan 606.

Key actions: Led a multidisciplinary research team that systematically tested hundreds of chemical compounds to find a synthetic dye that would target and destroy specific bacteria inside the body.

Significance: Developed Salvarsan 606 in 1909. This was the world's first synthetic chemical "magic bullet", which successfully cured syphilis.

Limitations: Salvarsan 606 was highly toxic because it was arsenic-based; it could easily kill the patient if administered in the wrong dosage. It was also highly specific, capable only of targeting the syphilis bacterium rather than treating a wide range of bacterial infections.

Gerhard Domagk — German Pathologist

Discovered the second magic bullet in 1932. He found that a red chemical dye called Prontosil effectively cured blood poisoning (streptococcus) in mice. He later bravely used it to successfully cure his own daughter's severe infection, proving its efficacy in humans.

Key actions: Discovered Prontosil in 1932, a sulfonamide compound that successfully killed streptococcal bacteria inside the blood of living mice.

Significance: Developed the second magic bullet and the first commercially available sulfonamide drug, providing a highly effective cure for deadly systemic infections like puerperal fever, pneumonia, and scarlet fever.

Limitations: Prontosil was a bacteriostatic sulfonamide, meaning it stopped bacteria from multiplying but did not actively destroy them, relying instead on the patient's own immune system to clear the infection. It was entirely ineffective against deep, necrotic tissue wounds (which were common on the Western Front) and could cause severe kidney damage.

Advances in Medicines: Magic Bullets and Antibiotics In 1909, Paul Ehrlich's team discovered Salvarsan 606, the first 'magic bullet'. It was a highly toxic, arsenic-based chemical compound that specifically targeted and destroyed the syphilis microbe inside the body without killing the rest of the patient's cells (Common Misconception: never call Salvarsan an antibiotic!). In 1932, Gerhard Domagk discovered a second magic bullet, Prontosil, which cured some types of blood poisoning. Scientists realised the active ingredient was sulphonamide, paving the way for targeted chemotherapy and cures for pneumonia and scarlet fever. The first true antibiotic (created using microorganisms rather than chemicals) was penicillin,

discovered by Alexander Fleming in 1928 and developed into a mass-produced drug by Florey and Chain in the 1940s. The development of magic bullets and antibiotics revolutionised medicine because, for the first time, doctors possessed targeted drugs that actively hunted down and killed deadly bacteria inside the human body.

KNOWLEDGE CHECK (Q1)

What is the key scientific difference between an antiseptic like carbolic acid and a magic bullet like Salvarsan 606?

- ☐ A. Antiseptics are used on the outside of the body to prevent infection, whereas magic bullets are injected into the body to target specific microbes.
- ☐ B. Antiseptics are created from microorganisms, whereas magic bullets are synthetic chemicals.
- ☐ C. Antiseptics are used to cure viruses, whereas magic bullets cure bacteria.
- ☐ D. There is no difference; they are both the same thing.

Q2. Identify the first 'Magic Bullet' discovered in 1909.

Alexander Fleming — Scottish Bacteriologist

Accidentally discovered Penicillin in 1928 after returning from a holiday to find that a mysterious mould (*Penicillium notatum*) had grown in a neglected petri dish and killed the surrounding staphylococcus bacteria.

Key actions: Left a petri dish of staphylococcus bacteria open on his laboratory bench before going on holiday in 1928, returning to find a mould (*Penicillium notatum*) had contaminated the dish and destroyed the surrounding bacteria.

Significance: Discovered the antibacterial properties of penicillin and published his findings in 1929. His paper served as the scientific starting point for subsequent antibiotic research.

Limitations: He was unable to extract, purify, or stabilize the active penicillin compound. Believing it was too unstable to be used as anything more than a minor local antiseptic, he abandoned his research in 1931, leaving it completely undeveloped as a systemic drug.

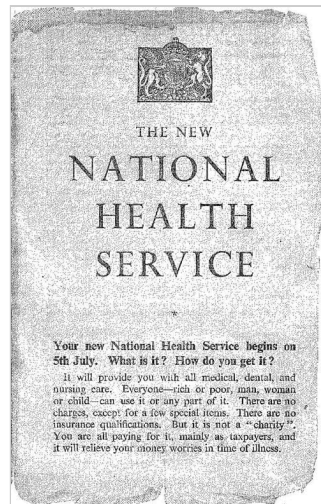
Howard Florey & Ernst Chain — Pharmacologist & Biochemist

Co-leaders of the Oxford University research team that successfully developed Penicillin into a usable medical drug during WWII. Florey travelled to the USA in 1941 to secure funding and industrial resources, while Chain discovered the complex chemical method to extract and purify Penicillin from the mould.

Key actions: Revived Fleming's research at Oxford University in 1938-1939; assembled a specialist team of biochemists and pathologists to purify, stabilize, and test penicillin on mice and humans.

Significance: Successfully developed penicillin into the world's first stable, usable, and systemic antibiotic by 1940-1941.

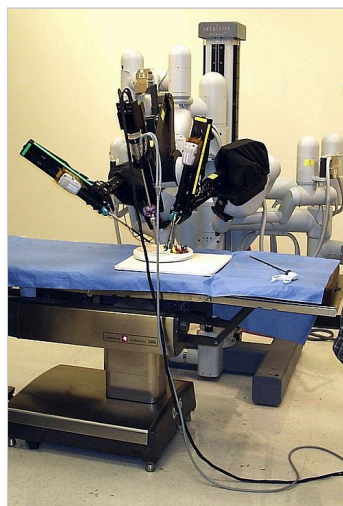
Limitations: They lacked the massive funding, industrial equipment, and factory space needed to mass-produce the drug in wartime Britain. Early patients (such as Albert Alexander in 1941) died when their supply of penicillin ran out, proving that scientific success was useless without industrial mass production.



Source B: Source B: 1948 Ministry of Health NHS Information Leaflet

Improved Access to Care: The Impact of the NHS Before the 20th century, seeing a doctor was too expensive for the poorest members of society. In 1942, the Beveridge Report called for a system that looked after people "from the cradle to the grave". In 1948, Minister of Health Aneurin Bevan launched the National Health Service (NHS). It provided free healthcare at the point of delivery, paid for by national taxes. The NHS brought hospitals, GPs (General Practitioners), dentists, and maternity care under one massive national organisation, eventually leading to a massive increase in the nation's life expectancy. The NHS fundamentally transformed British society because it ensured that every single citizen finally had equal access to vital medical diagnosis and treatment, entirely removing the fear that getting sick would bankrupt a family.

Q3. Describe how Magic Bullets work.



Source C: Source C: Modern Da Vinci Robotic Keyhole Surgery Unit

High-Tech Medical and Surgical Treatment With pain and infection largely solved in the 19th century, modern surgery became highly advanced. The first successful kidney transplant took place in 1954, followed by the first heart transplant in 1967. Keyhole surgery (laparoscopic surgery) and microsurgery use tiny cameras and narrow surgical instruments to operate inside the body through tiny incisions, drastically reducing recovery times and trauma. Other high-tech hospital treatments include radiotherapy and chemotherapy to shrink and destroy cancer cells, and dialysis machines to wash the blood of

patients with kidney failure. The rapid development of high-tech treatments turned previously fatal internal conditions and organ failures into manageable problems, drastically extending the lives of severely ill patients.

Q4. Explain how diagnosing illness changed in the 20th century.

New Approaches to Prevention: Mass Vaccinations and Lifestyle Campaigns Building on Pasteur and Koch's work, the government funded mass vaccination campaigns to wipe out deadly childhood diseases, including diphtheria (1942), polio (1956), and measles (1968). Because lifestyle factors like smoking and poor diet were proven to cause illnesses like cancer and heart disease, the government shifted from a 'laissez-faire' attitude to active intervention. The government uses 'nudge tactics' (like the Sugar Tax) and laws (the 2007 smoking ban in public places) to force healthier choices. They also run major advertising campaigns, such as Change4Life (encouraging families to eat less sugar and exercise more) and Stoptober (encouraging smokers to quit). These new approaches shifted the government's focus away from just treating sick patients and towards actively educating the public to prevent deadly diseases from occurring in the first place.

Q5. To what extent did the discovery of Magic Bullets revolutionize treatment?

Pair & Share Activity

Q6. Which development was the most significant turning point in twentieth-century healthcare: the scientific breakthrough of antibiotics, or the political creation of the National Health Service (NHS)?

GCSE Exam Practice

Q7. Explain why the government took a more active role in preventing disease in the period c1900-present. (12 marks)

Q8. Explain one way in which ideas about the prevention of illness in the modern period (c1900-present) were different from the medieval period (c1250-c1500). (4 marks)

L3: KT4.3: How was Penicillin discovered and mass-produced?

LEARNING OBJECTIVES

- ☐ Contrast the roles of Alexander Fleming with those of Howard Florey and Ernst Chain in the penicillin breakthrough.
- ☐ Analyze how the factors of government intervention, science and technology, and war collaborated to enable the mass production of penicillin.

Before the 20th century, a simple scratch or an infected wound could easily lead to fatal blood poisoning. The discovery of penicillin, the world's very first antibiotic, marked a monumental turning point in modern medicine. Driven by the desperate, bloody necessity of the Second World War, scientists finally managed to harness naturally occurring microbes to actively destroy bacterial infections, saving millions of lives and permanently transforming modern healthcare.

Did you know?

- 1928: The exact year Alexander Fleming accidentally discovered penicillin mould killing bacteria in his unwashed petri dish.
- 1941: The year Florey and Chain successfully treated their first human patient, Albert Alexander, proving the purified drug worked.
- 15%: The estimated percentage of injured Allied soldiers whose lives were saved by mass-produced penicillin during WWII.

Recall & Retrieval

- | | |
|--|---|
| 1. What is a 'magic bullet'? | 2. Name the first magic bullet, discovered by Paul Ehrlich in 1909. |
| 3. What was established in Britain in 1948 to provide free medical care? | 4. Who was the Minister of Health responsible for establishing the NHS in 1948? |
| 5. Name the second magic bullet, discovered by Gerhard Domagk in 1932 to treat strep infections. | 6. What structure did Watson and Crick discover in 1953? |
| 7. Who published the Germ Theory in 1861? | 8. Who introduced antiseptic surgery using carbolic acid in 1865? |
| 9. Name one modern lifestyle factor that causes disease. | 10. Which female scientist's X-ray crystallography photo was vital to Watson and Crick's DNA discovery? |

Vocabulary Check

Conceptual Classification: Categorise the terms from the word bank into the two historical boxes below:

Penicillin | Alexander Fleming | Florey and Chain | Culture Medium | Mass Production | Antibiotic Resistance

Power, Governance & Warfare

Economy, Trade & Society

Source S1 : Alexander Fleming's Contaminated Petri Dish (1928)



*The famous culture plate showing the fungal colony of *Penicillium notatum* surrounded by a clear 'halo' where *staphylococcus* bacteria were destroyed.*

Source S2 : Penicillin Culture Flasks at Wartime Production Plant (1943)



A laboratory technician checking rows of thousands of culture flasks growing penicillin mould on nutrient liquid during wartime mass production.

Source S3 : US War Production Board WWII Penicillin Poster (1944)



'Thanks to PENICILLIN... He Will Come Home!': US wartime poster highlighting penicillin's role in slashing soldier deaths from wound infections.

The Miracle of Penicillin In 1928, Alexander Fleming left a petri dish of staphylococcus bacteria out while on holiday. When he returned, he noticed a mould (Penicillium) growing on it, which was actively killing the bacteria. He had accidentally discovered the world's first antibiotic. However, Fleming couldn't extract enough pure penicillin to use as a medicine, so he simply wrote a paper on it and gave up. Ten years later, two scientists named Florey and Chain read Fleming's paper. With funding from the US government during World War II, they developed a way to mass-produce the drug. By 1944, there was enough Penicillin to treat all the Allied casualties on D-Day. After the war, mass production techniques were shared globally, making antibiotics the most widely used medical treatment in history.



Source A: *Source A: Alexander Fleming's Contaminated Petri Dish (1928)*

Alexander Fleming's Accidental Discovery (1928) In 1928, the Scottish scientist Alexander Fleming went on holiday and accidentally left a stack of petri dishes containing staphylococcus bacteria unwashed on his laboratory bench at St Mary's Hospital, London. Upon returning, he noticed a spore of mould had blown into a dish and grown, creating a clear, bacteria-free zone around itself. He identified this bacteria-killing mould as *Penicillium notatum*. Fleming published his findings in a medical journal in 1929, correctly stating that the mould juice killed harmful bacteria. However, he lacked the funding and advanced chemical equipment required to extract pure penicillin. He lost interest, and his discovery was largely ignored for a decade. Fleming's accidental observation was a crucial first step, but because he could not purify it or perform trials on infected humans, penicillin remained just an interesting laboratory observation rather than a usable drug.

Q1. GCSE Analytical Assessment: Evaluate the miraculous antibacterial breakthrough of penicillin against its industrial challenges and modern limitations.



Source B: Source B: Penicillin Culture Flasks at Wartime Production Plant (1943)

The Brilliant Teamwork of Florey and Chain (1939-1941) In 1939, a highly skilled team of scientists at Oxford University, led by Howard Florey and Ernst Chain, read Fleming's forgotten article and decided to investigate further. Using advanced freeze-drying techniques and complex chemical processes, Chain successfully managed to extract a tiny amount of pure penicillin powder. In 1940, they successfully cured mice that had been deliberately infected with deadly streptococcus bacteria. In 1941, they conducted their first human trial on a dying policeman named Albert Alexander, who had contracted a severe infection from a rose thorn scratch. He recovered miraculously within days, but tragically died when their tiny supply of the drug ran out. Florey and Chain's collaborative research proved undeniably that penicillin could safely cure life-threatening bacterial infections inside the human body without harming the patient.

Q2. Describe how Penicillin was discovered.



Source C: Source C: US War Production Board WWII Penicillin Poster (1944)

Mass Production and the Second World War Despite proving its incredible potential, Florey could not persuade British chemical companies to mass-produce the drug because they were entirely focused on making explosives for the Second World War. In July 1941, Florey travelled to the USA to persuade the

American government to invest. Once the USA joined the war in December 1941, the government gave massive interest-free loans to huge chemical companies to build colossal factories, pioneering the breakthrough 'deep-fermentation tank' method to grow the mold on an industrial scale. By 1943, mass production was fully underway. By D-Day (June 1944), enough penicillin had been manufactured using these deep-fermentation tanks to treat every single infected Allied casualty, eventually saving an estimated 15% of injured soldiers from dying of gangrene or blood poisoning. The desperation of the Second World War finally forced governments to pour massive financial investment into mass-producing penicillin. (Common Misconception: Students frequently state that Fleming mass-produced penicillin during WWI. You must remember the strict chronology: Fleming made the accidental discovery in 1928, it remained a laboratory curiosity until Florey and Chain purified it in 1940, and mass production only scaled up during WWII in 1943-44.)

KNOWLEDGE CHECK (Q3)

What was the main scientific and practical limitation that caused Fleming to abandon his work on penicillin in 1931?

- ☐ A. He realized that penicillin was highly toxic to humans.
- ☐ B. He could not secure funding from the government because it was peacetime.
- ☐ C. He lacked the chemical expertise and facilities to extract and purify penicillin in large enough quantities for human trials.
- ☐ D. He was drafted into the army to fight in World War II.

Q4. Explain the roles of Florey and Chain in the development of Penicillin.

Q5. Evaluate the role of World War II in the mass production of Penicillin.

Pair & Share Activity

Q6. Who deserves the most credit for the penicillin breakthrough: Alexander Fleming for his accidental discovery in 1928, or Howard Florey and Ernst Chain for their scientific development and therapeutic isolation of the drug in the 1940s?

GCSE Exam Practice

Q7. Explain why there was rapid progress in the development of penicillin in the 1940s. (12 marks)

Q8. Explain one way in which the treatment of illness in the modern period (c1900-present) was different from the treatment of illness in the period c1700-c1900. (4 marks)

L4: KT4.4: How has the government tackled the modern epidemic of Lung Cancer?

LEARNING OBJECTIVES

- ☐ Explain how modern science and technology have transformed the diagnosis and treatment of lung cancer since c1900.
- ☐ Analyze the effectiveness and changing focus of government legislation to prevent smoking-related lung cancer.

Lung cancer was incredibly rare in the nineteenth century but rapidly became the second most common cancer in the UK by the twenty-first century. After the link between smoking and cancer was conclusively proven in 1950, medicine had to develop advanced high-tech methods to diagnose and treat this deadly disease. Simultaneously, the crisis forced the government to abandon its reluctance to interfere in personal habits, leading to aggressive modern prevention campaigns.

Did you know?

- 1950: The year the British Medical Research Council published its conclusive study linking cigarette smoking to lung cancer.
- 2007: The year the government passed a law banning smoking in all enclosed public workplaces and raised the legal age for buying tobacco to 18.
- 85%: The approximate percentage of lung cancer cases that are caused by people who smoke or have previously smoked.

Recall & Retrieval

- | | |
|---|--|
| 1. Who discovered Penicillin by accident in 1928? | 2. Which two scientists at Oxford developed Penicillin into a usable drug? |
| 3. What major global event drove the mass production of Penicillin? | 4. What is a 'magic bullet'? |
| 5. What was established in Britain in 1948 to provide free medical care? | 6. Who was the Minister of Health responsible for establishing the NHS in 1948? |
| 7. What are the four humours? | 8. Who wrote 'On the Fabric of the Human Body' (1543)? |
| 9. Why is lung cancer so difficult to treat effectively compared to other diseases? | 10. Which landmark 1950 epidemiological study proved the link between smoking and lung cancer? |

Vocabulary Check

Spot the Deliberate Error: Read the statement below. One historical fact or vocabulary concept has been deliberately falsified. Underline the error and explain the accurate historical reality below:

Carcinogen | Bronchoscopy | CT Scanner | Radiotherapy | Immunotherapy | Nanny State

Challenge: Write ONE statement containing a deliberate historical misconception using **Carcinogen** or **Bronchoscopy**. Identify and correct the error:

Teacher Model / Historical Correction:

Source S1 : 1949 "More Doctors Smoke Camels" Print Advertisement

More people are smoking **CAMELS** than ever before!

YOUR "T-ZONE"
WILL TELL YOU...
T for Taste...
T for Throat...
That's your proving ground
for any cigarette. See if
Camels don't suit your
"T-Zone" to a T.

BALLET STAR Kathryn Lee has her own reasons...from her own experience: "During the wartime cigarette shortage, I tried many different brands," says Miss Lee. "I compared them... and learned by experience that Camels suit my 'T-Zone' to a 'T.' I've smoked Camels ever since!"

Thousands and thousands of smokers had the same experience. They compared—discovered the differences in cigarette quality...found that cool, mild, full-flavored Camels suit them best.

Try Camels on your "T-Zone." Let your own experience tell you why more people are smoking Camels than ever before!

According to a Nationwide survey:

MORE DOCTORS SMOKE CAMELS THAN ANY OTHER CIGARETTE

When 313,347 doctors were asked by three independent research organizations to name the cigarette they smoked, more doctors named Camel than any other brand!

'According to a Nationwide survey: MORE DOCTORS SMOKE CAMELS THAN ANY OTHER CIGARETTE': mid-century tobacco advertisement using medical endorsement.

Source S2 : NHS Anti-Smoking Public Health Education Campaign



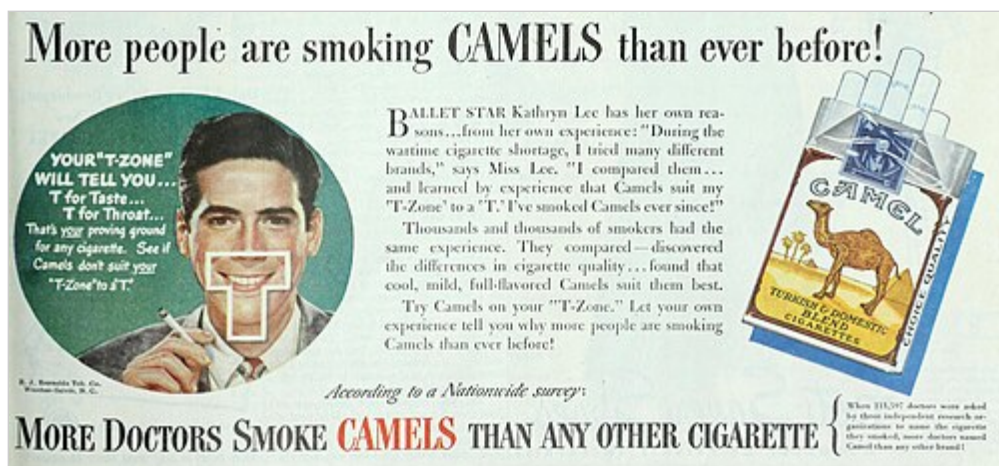
Public education anti-smoking campaign illustrating the direct link between cigarette smoke inhalation, tar deposits, and cancerous lung tumors.

Source S3 : Standardised Plain Packaging Display (Children & Families Act)



Retail display of standardized plain cigarette packs: all brand logos and promotional fonts stripped away, replaced by graphic health warnings.

Tackling Lung Cancer By the mid-20th century, lung cancer cases had skyrocketed, primarily due to the massive increase in cigarette smoking. Unlike infectious diseases, lung cancer could not be cured with antibiotics. Diagnosing it requires advanced technology like CT scans and PET scans, and treatments are aggressive, involving surgery to remove tumours, radiotherapy (using radiation to kill cancer cells), and chemotherapy. Because treatments are often ineffective if the cancer is caught late, the UK government focused heavily on prevention. In 2007, smoking was banned in all enclosed public workspaces. In 2015, plain packaging laws were introduced, making cigarette boxes look unappealing. Furthermore, massive public health campaigns and taxation have been used to persuade people to quit, showing a huge shift in government involvement in public health.



Source A: Source A: 1949 "More Doctors Smoke Camels" Print Advertisement

The Rise of Lung Cancer and Early Diagnosis In 1950, the British Medical Research Council conclusively proved that the massive rise in lung cancer cases was linked to smoking cigarettes. Early diagnosis was very difficult; doctors initially relied on traditional x-rays, but these were not detailed enough and often mistook cancer for other minor lung conditions like abscesses. Because early diagnosis using basic x-rays was so ineffective, the disease was usually only discovered when it was too late to cure, forcing scientists to develop much more advanced imaging technology.

Q1. Identify one method used to diagnose Lung Cancer.

High-Tech Diagnosis Today, patients are initially given a CT scan, which creates a highly detailed 3D picture of the inside of the lungs (often using injected dye to make the lungs show up clearly). If a tumour is spotted, doctors use a PET-CT scan (injecting a small amount of radioactive material) to accurately identify active cancerous cells. Doctors will also use a bronchoscope (a flexible camera passed down into the lungs) to collect a direct sample of the cells, known as a biopsy. This high-tech combination of scans and biopsies allows doctors to determine the exact type and stage of the cancer, enabling them to create a highly accurate, targeted treatment plan for the patient.

Q2. Describe the aggressive treatments used for Lung Cancer.



Source B: Source B: NHS Anti-Smoking Public Health Education Campaign

Science and Technology in Treatment If diagnosed early, surgeons can operate to physically remove the tumour or even the entire infected lung. Radiotherapy uses concentrated beams of radiation aimed directly at the tumour to shrink it and prevent it from growing. Chemotherapy involves injecting powerful chemical drugs into the bloodstream to shrink the tumour before surgery or destroy remaining cancer cells afterwards. While there is still no guaranteed cure for advanced lung cancer, these powerful scientific treatments can actively shrink tumours and destroy cancerous cells, significantly prolonging the lives of patients.

Q3. Explain why the UK government has focused heavily on preventing Lung Cancer.



Government Action and Prevention Initially, the government was slow to act because they earned around £4 billion from tobacco taxes. However, rising death rates forced them to intervene by influencing and changing behaviour. To influence behaviour, the government banned cigarette television adverts in 1965 and completely banned all tobacco advertising and sports sponsorship by 2005. They also frequently increased the tax on tobacco. To force a change in behaviour, the government passed a landmark public health law in 2007 banning smoking in all enclosed public workplaces, and raised the legal age to buy tobacco from 16 to 18. In 2012, they made it illegal for shops to openly display cigarettes. This aggressive government action represented the ultimate transition to a modern regulatory state, proving they were finally willing to actively restrict personal freedoms and attack highly profitable businesses to protect the public and prevent modern lifestyle diseases.

KNOWLEDGE CHECK (Q4)

What is the key clinical advantage of using a CT scan over a traditional chest X-ray when diagnosing a lung cancer patient?

- ☐ A. A CT scan uses harmless sound waves instead of radiation.
- ☐ B. A CT scan can be performed at home by the patient.
- ☐ C. A CT scan creates highly detailed, 3D cross-sectional images of the lungs, allowing doctors to detect much smaller tumors than a 2D X-ray.
- ☐ D. A CT scan can actively cure the lung cancer by destroying the tumor.

Q5. Evaluate the effectiveness of government legislation in reducing smoking.

Pair & Share Activity

Q6. Which category of modern government action has been the most significant turning point in the prevention of lung cancer: directly warning and educating current smokers (such as plain packaging and graphic warning labels), preventing young people from starting (such as raising the legal age of sale from 16 to 18), or protecting non-smokers from second-hand smoke (such as the 2007 indoor smoking ban)?

GCSE Exam Practice

Q7. Explain how science and technology have improved the diagnosis and treatment of lung cancer. (12 marks)

Q8. Explain one way in which hospital care in the modern period (c1900-present) was different from hospital care in the period c1700-c1900. (4 marks)

